

Information Management

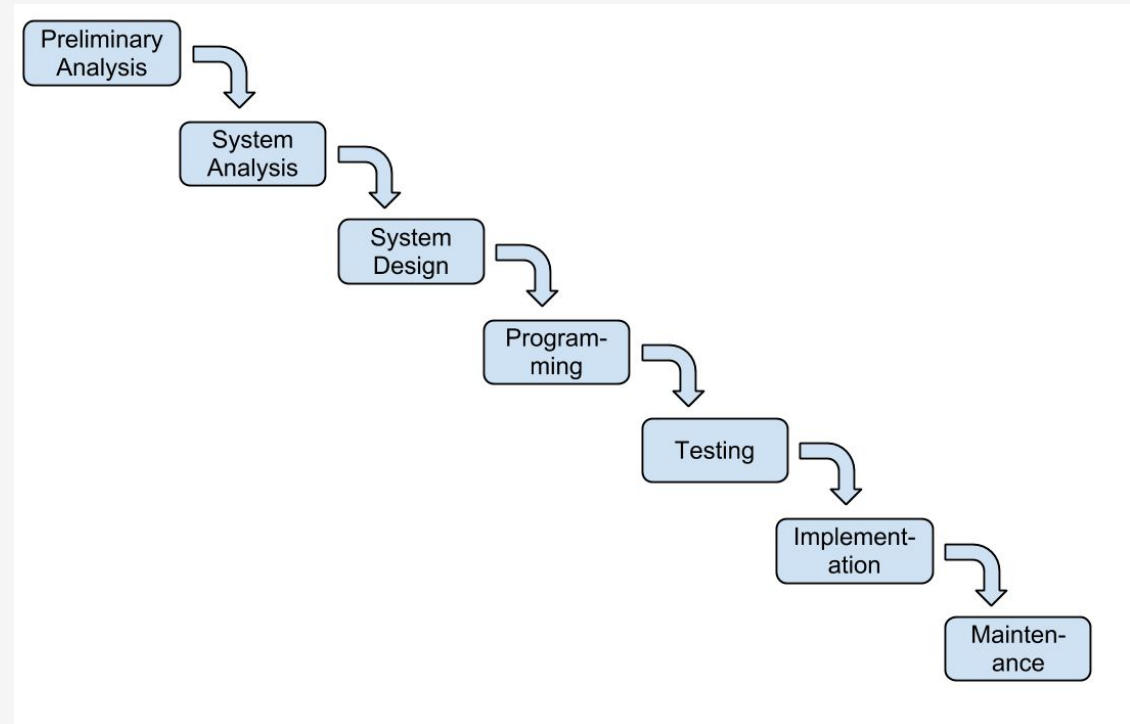
System Development Life Cycle

Introduction

The systems development life cycle (SDLC), also referred to as the application development life-cycle, is a term used in systems engineering, information systems and software engineering to describe a process for planning, creating, testing, and deploying an information system.

System Development Life Cycle

The first development methodology we are going to review is the systems-development life cycle (SDLC). This methodology was first developed in the 1960s to manage the large software projects associated with corporate systems running on mainframes.

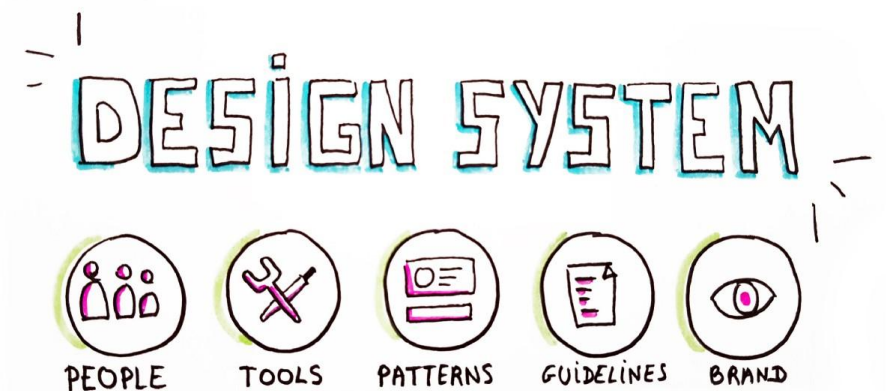


System Development Life Cycle (Conti...)

1. Preliminary Analysis - In this phase, a review is done of the request. Is creating a solution possible? What alternatives exist? What is currently being done about it? Is this project a good fit for our organization?

2. System Analysis - In this phase, one or more system analysts work with different stakeholder groups to determine the specific requirements for the new system. No programming is done in this step.

3. System Design - In this phase, a designer takes the system-requirements document created in the previous phase and develops the specific technical details required for the system. It is in this phase that the business requirements are translated into specific technical requirements.



System Development Life Cycle (Conti...)

4. Programming - The code finally gets written in the programming phase. Using the system-design document as a guide, a programmer (or team of programmers) develops the program.

5. Testing - In the testing phase, the software program developed in the previous phase is put through a series of structured tests. The first is a unit test, which tests individual parts of the code for errors or bugs.

6. Implementation - Once the new system is developed and tested, it has to be implemented in the organization. This phase includes training the users, providing documentation, and conversion from any previous system to the new system.



System Development Life Cycle (Conti...)

7. Maintenance – This final phase takes place once the implementation phase is complete. In this phase, the system has a structured support process in place: reported bugs are fixed and requests for new features are evaluated and implemented; system updates and backups are performed on a regular basis.



System Life Cycle – Stages and Activities

The development of a system and its place within the organization, knowledge of the life cycle of systems is necessary. Like human beings, all systems go through a life cycle. Systems originate, develop, and, finally, decline. Several models have been developed that illustrate the concept of a system life cycle. In this section, four perspectives of system life cycles.

- General system life cycle
- Information system life cycle
- Discontinuity of information system life cycle
- Organization-wide information system life cycle



Information systems are similar to biological, social, and political systems. All systems go through a system life cycle. All systems have a birth or development period, a period of growth, a period of maturity, and a period of decline or deterioration.

The following example demonstrates the four phases of the general system life cycle.

Birth and development (phase 1)

Growth and maturity (phases 2 and 3)

Decline (phase 4)

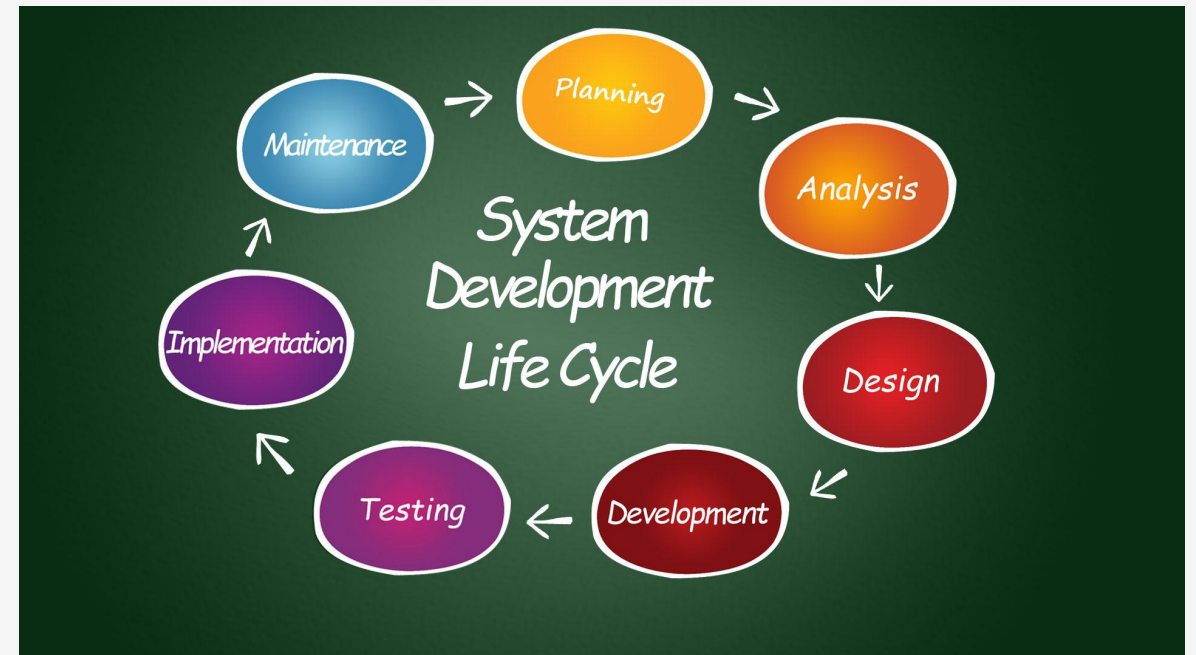


GSLC
(General System
Life Cycle)

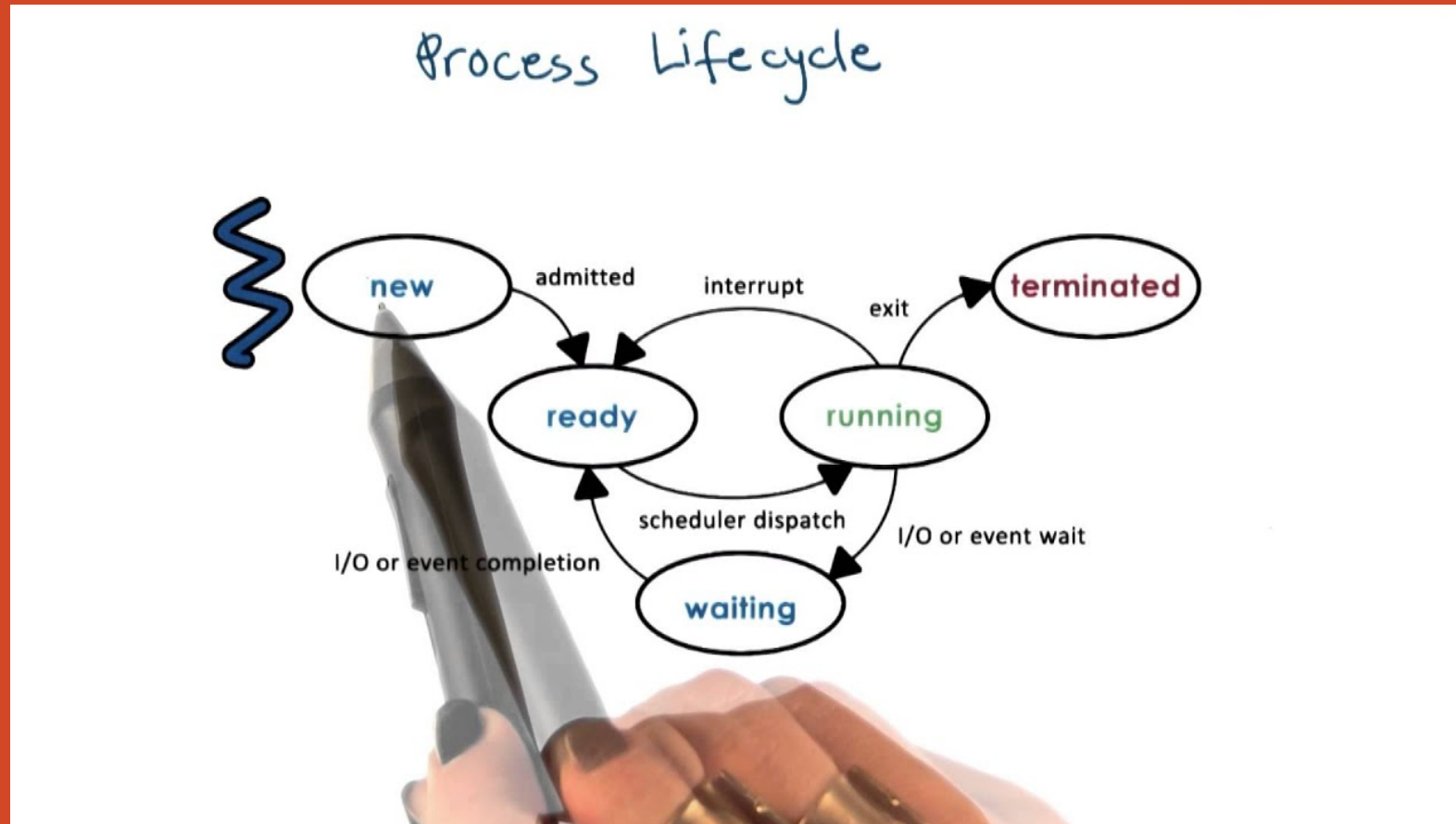
Main System Development Life Cycle Phases

As we covered before, the SDLC is used as a conceptual model that includes the procedures and policies necessary to develop or alter a system throughout a life cycle. The end result should be a high-quality system that meets or exceeds customer expectations and is within time and budget constraints. Within this framework, these are the main system development life cycle phases:

- Planning
- Analysis
- Design
- Development
- Testing and integration
- Implementation
- Maintenance



Process Life Cycle



Information Processing Cycle

Categorizing and clustering data is an integral part of information processing. Information processing is a term used to describe the analysis of data, with the goal of extracting meaning from it. The main function of the body's information processing system is to maintain homeostasis (homeostasis is defined as balanced environment, or a state of health).

What is Information Processing?

Information processing is the process of changing or converting information into meaningful information. It is processed, organized or classified data which is useful for the receiver. Information is the processed data which may be used “as it is” or may be put to use along with more data or information. The receiver of information takes actions and decisions based on the information received.

Further information is useful & meaningful only if it has these characteristics:

Timely – Information should be available when required, a delay in obtaining information renders it useless.

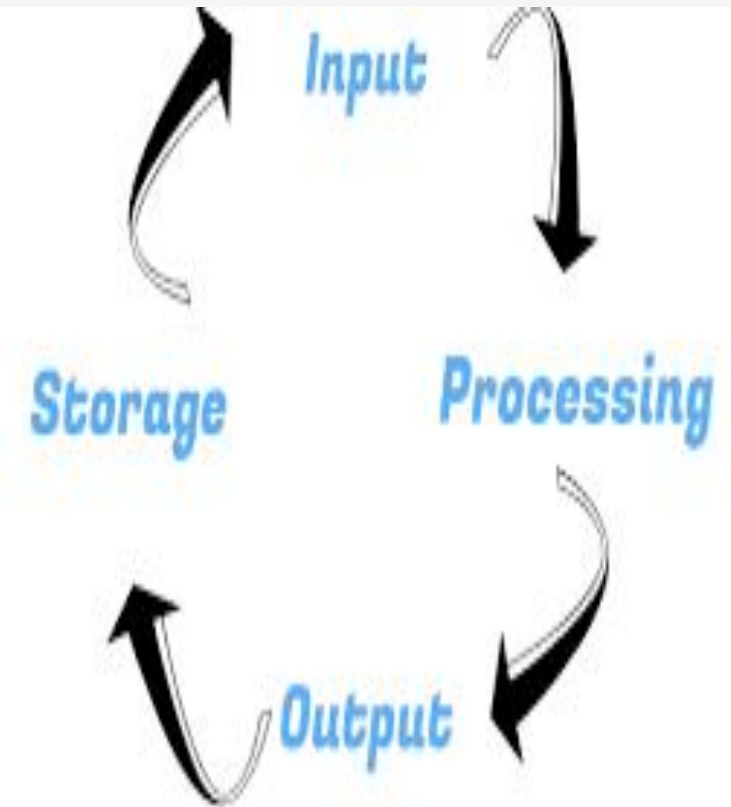
Accuracy – Accuracy of information has a significant impact on the decision-making. Possibilities of even slightest errors should be minimized.

Completeness – Information should be complete. Incomplete information causes incorrect and unintended results.

Comprehensive – Information which is incomprehensible is useless for the receiver. This becomes a case of information failure as the sender sent the information, but it was of no use for the receiver, thus is not considered as “information.”

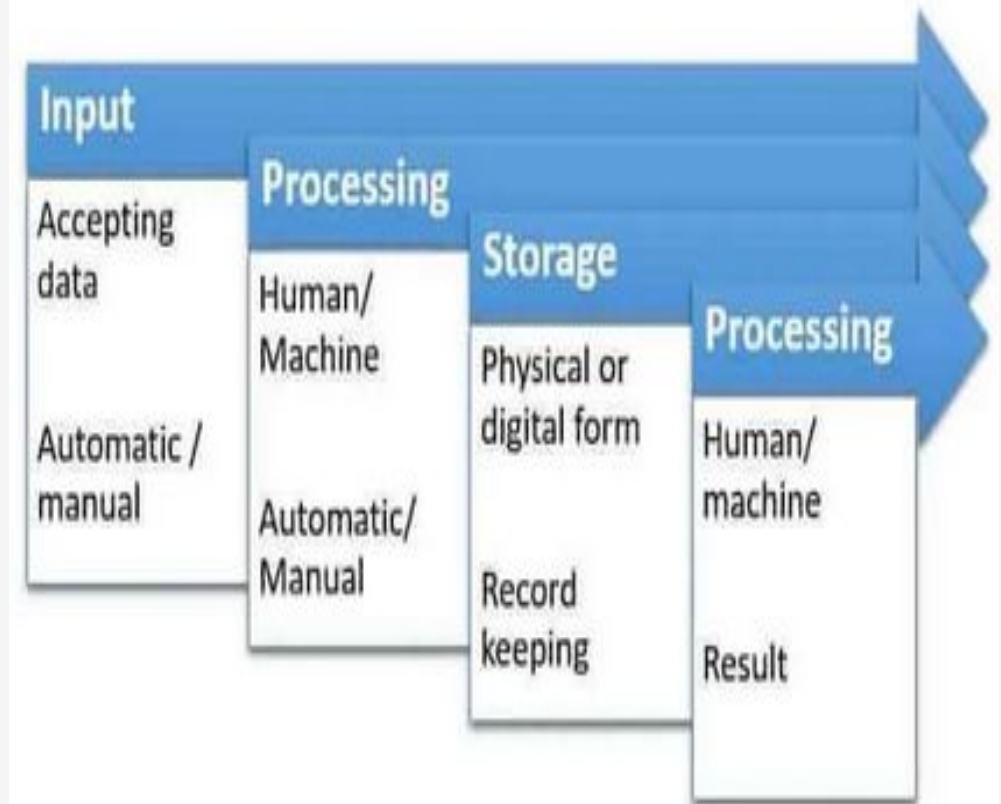
Understanding Information Processing Cycle

Information processing cycle is a sequence of events consisting of Input, Processing, Storage & Output. To understand more about what is information processing cycle it is a good idea to study about data processing cycle also. These events are similar as in case of data processing cycle. For a computer to perform useful work, the computer has to receive instructions and data from the outside world. The computer receives data and instructions during the INPUT stage of the information processing cycle. Useful information results are obtained when appropriate inputs are applied to data. Applying instructions to data takes place during the PROCESSING stage of the information processing cycle. To avoid having to re-enter data and instructions or reprocess information, computers can save information. Saving information on a computer occurs during the STORAGE phase of the information processing cycle.



Four Phases of the information processing cycle

1. **Input:** Computer receives data and instructions.
2. **Process:** Computer applies instructions to data to produce information (organized Data).
3. **Storage:** Saving the information for a subsequent use or use in future.
4. **Output:** Computer sends information to people in a usable format.



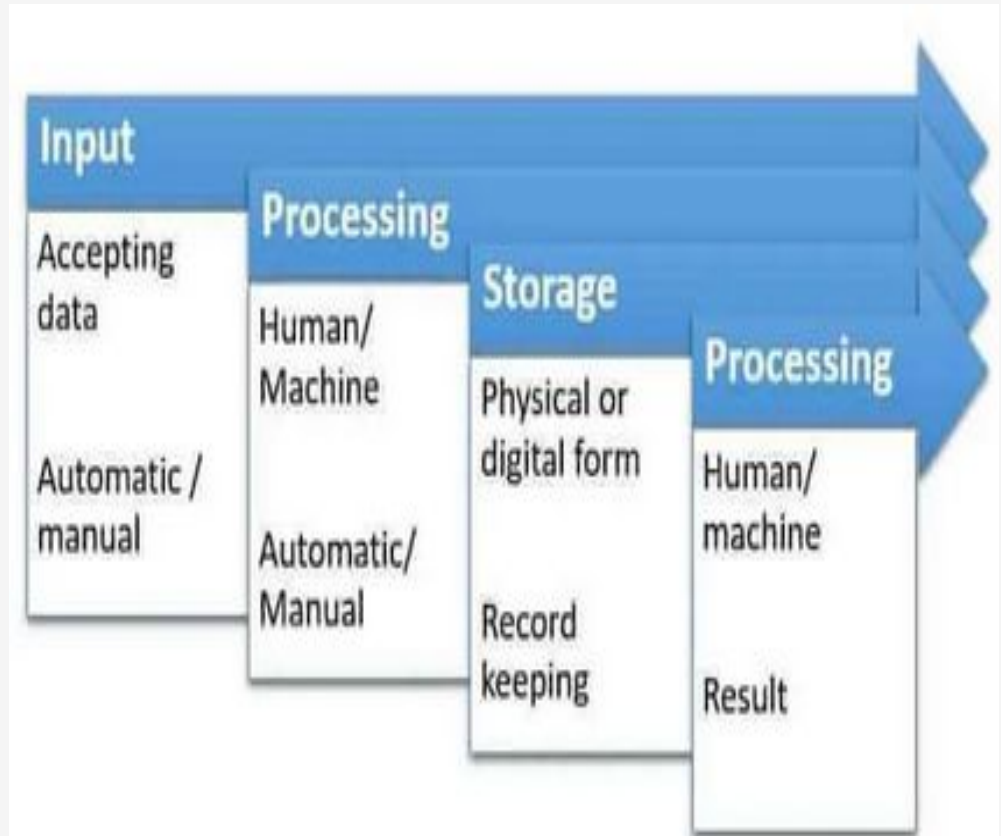
Four Phases of the information processing cycle

1. **INPUT** – Entering data into the computer

A. Feeding the collected raw data or data entry in the cycle for processing. This is the raw data which is supplied for processing & obtaining information.

B. Input can be done by utilizing various input devices such as keyboards, mice, flatbed scanners, barcode readers, joysticks, digital data tablets (for graphics drawing), electronic cash registers, etc.

C. Input process is most essential as any error in the input data or the input process will result in wrong output.



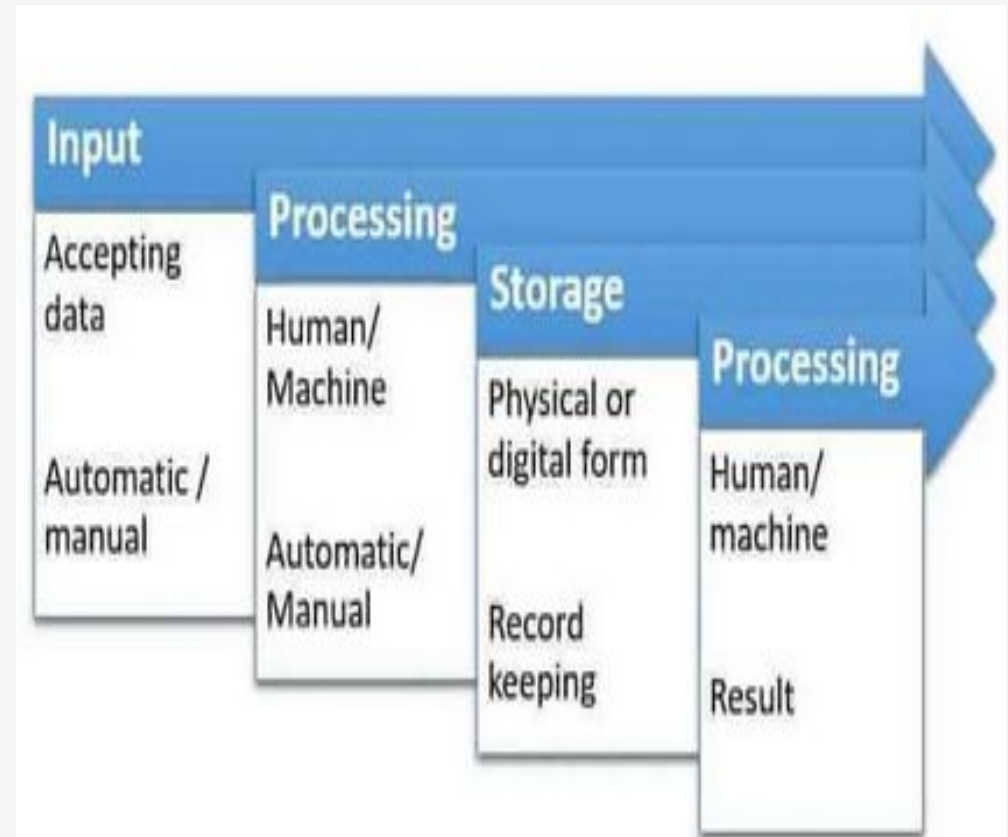
Four Phases of the information processing cycle

2. PROCESSING – Performing operations on the data

A. Once the input is provided the raw data is processed by a suitable or selected processing method. Processing information is the most crucial step as it allows the processed data in the form of output which will be used further.

B. Processing is usually done by CPU (Central Processing Unit) in a computer. CPU is the crucial component for getting the operations done. The speed of processing depends on the computing power and processing power of the processor.

C. Processing can be done by using various processing software's and computer system depending upon the requirement.

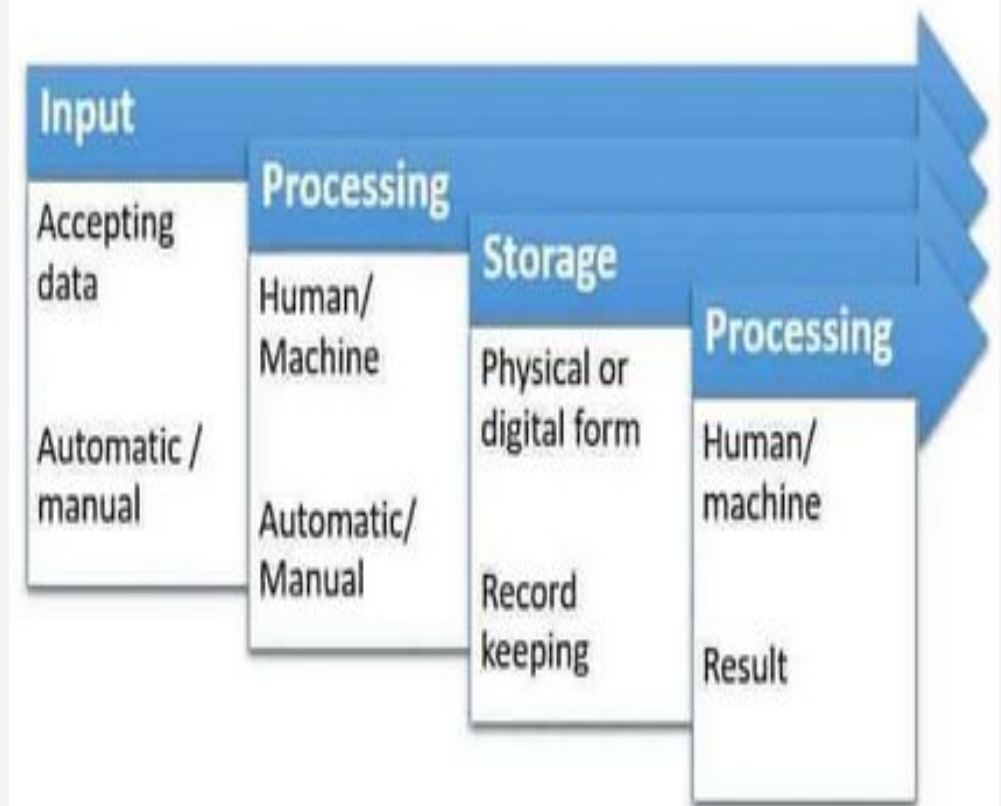


Four Phases of the information processing cycle

3. **STORAGE** – Saving data in a soft/physical form

A. The data which is to be fed and the output which will be generated needs to be stored. This stage is at times overlapping with the output stage since.

B. Storage can be done on various storage devices such as external hard disk, inbuilt hard disk, data centers, pen drives, micro SD cards, compact disks or even in registers.

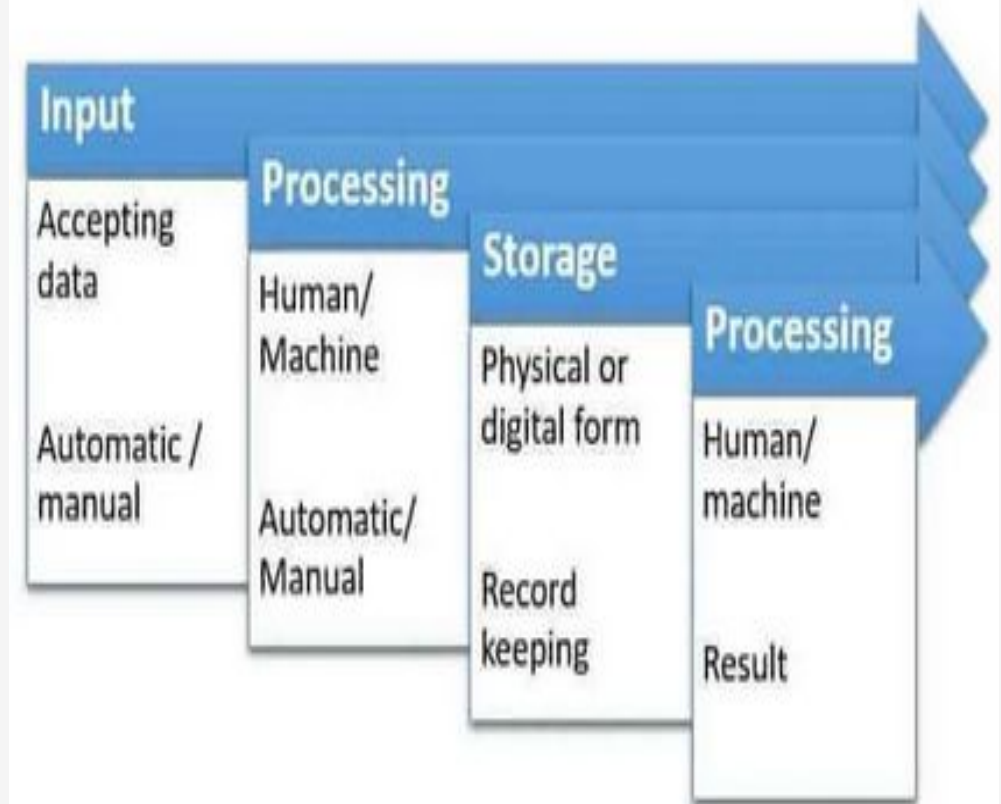


Four Phases of the information processing cycle

4. OUTPUT – Results obtained, i.e., information

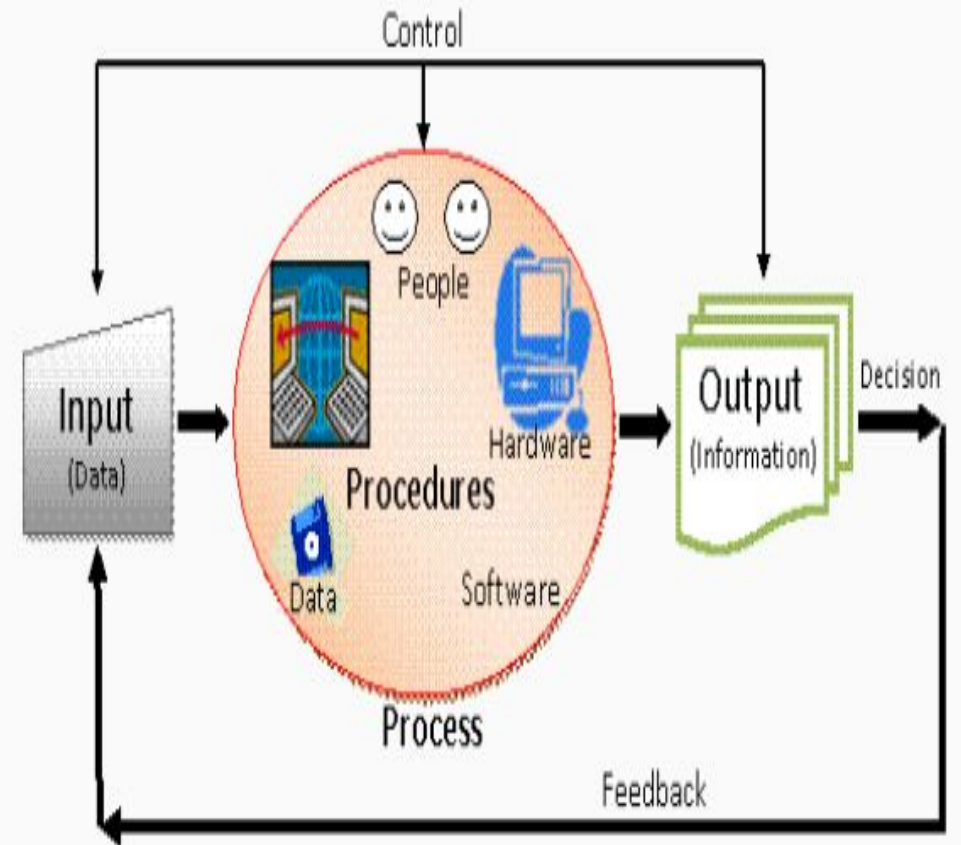
A. This is the outcome, and the raw data provided in the first stage is now “processed,” and the data is useful and provides information and no longer called Data. This might be further used for data visualization.

B. This can be used as it is or used for further processing along with more data. Output device can be a paper or in form of display screen such as monitors or phone screen.



System Development Approaches

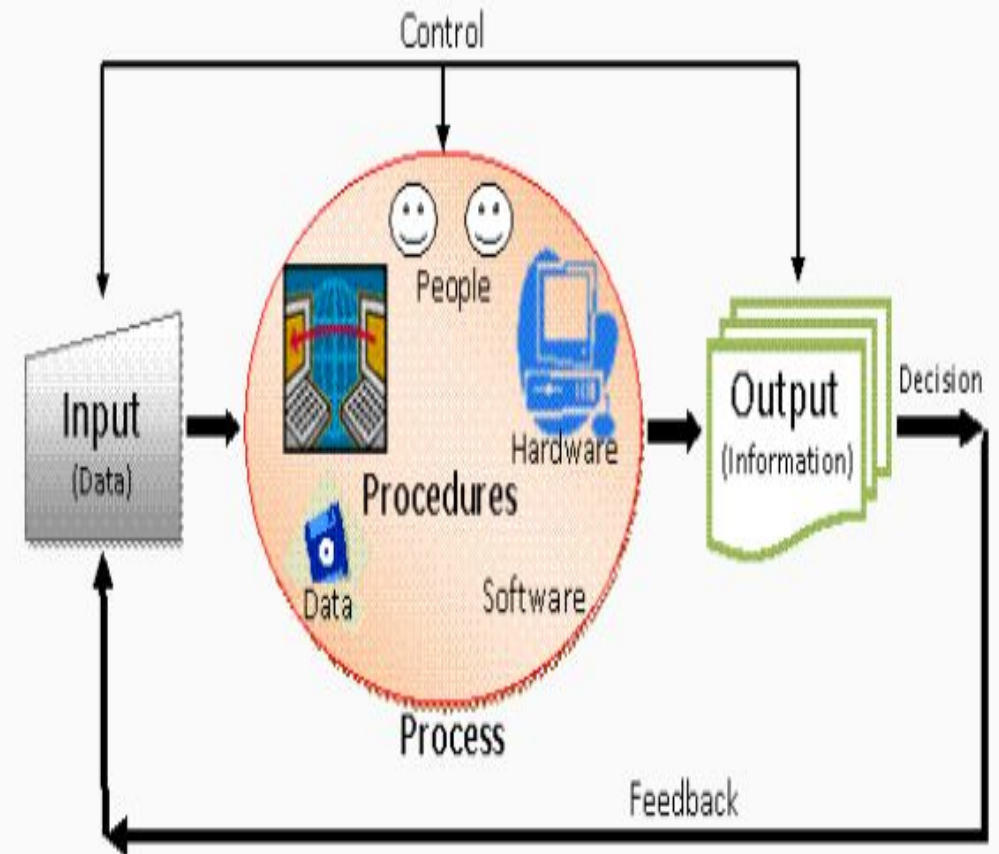
The word system is possibly the most overused word in our vocabulary and is used quite often in everyday life. Generally, we speak about political system, education system, human body system, social system, transportation system, banking system, computer system etc. The common feature, which all systems share is that they are collection of elements integrated to achieve the required goals. The term system is derived from Greek word system, which means an organized relationship among functioning units or components.



System Development Approaches

A system exists because it is designed to achieve one or more objectives. In present context, the word System can be defined as a group of interdependent components in an organized way, which works together to achieve one or more objectives as per specified plan. Interdependent components may refer to physical parts or managerial steps known as subsystem of a system. For example, a dairy plant can be considered as a system in which various departments and sections are joined together for a common goal.

A system may be regarded as a set of entities or elements interacting among themselves in a certain pre-specified manner to process inputs and produce outputs (objectives) in a defined time period. Output of a system may be quantitative or qualitative.



Characteristics of a System

As defined, a system is an organized relationship of various subsystems or components to achieve some objectives.

Thus the definition of a system suggests the following characteristics that are present in all systems.

- **Organization**
- **Interaction**
- **Interdependence**
- **Integration**
- **Planned approach**
- **Central objective**



THE END...



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